SKYFORCE

Remaining Simulation Issues

Technical briefing for guide discussion

Prepared 10 October 2026 | Based on the final-resolution PDF and latest MATLAB CSV export

Purpose

Summarise the specific items that still need a guide's decision or technical review before the MATLAB report is finalised. The revised resolution note closes several conceptual questions, but the latest simulation export and the resolution PDF are not yet fully synchronised. This brief asks for decisions, not invented performance guarantees.

Executive summary

Item	Current position	What we need from the guide
Mission energy	Latest phase-energy values differ from the values used in the resolution PDF.	Approve one consistent calculation basis and whether to run a continuous launch-to-landing energy model.
Closing speed	Target velocity components/headings are present in the latest CSV; interceptor velocity and line-of-sight components are not exported.	Confirm the closing-speed definition and review an independent numerical check.
Forward-flight propulsion	15-30% thrust de-rating is a sensitivity assumption, not measured data.	Confirm whether sensitivity analysis is sufficient for the hackathon report or whether a speed-dependent model is required.
Mass ledger / CAD	18.00 kg is represented as 14.78 kg subtotal plus 3.22 kg integration/growth allowance.	Confirm whether this allowance is acceptable at this stage or must be replaced by a component-level bill of materials before CAD freeze.
Capture/recovery	Net and parachute dimensions are sizing targets; deployment and capture performance remain unvalidated.	Confirm whether the stated scope/limitations satisfy the deliverable.
Site boundaries	300 m keep-out and 120 m safe radii are illustrative placeholders.	Confirm these are acceptable for a hypothetical scenario without an actual site layout.

Authoritative baseline assumed in this brief

Nominal mass: 18.00 kg; post-capture mass: 23.50 kg; T-Motor U8 Lite KV100; G28x9.2 CF 28-inch propeller; Alpha 60A 12S ESC; 7.3 kgf static thrust per motor; 25 m/s speed cap; 4.5 m/s^2 model acceleration cap; 207 Wh no-tow sizing ceiling and 253 Wh tow ceiling. The mass ledger currently assigns 3.22 kg to integration/growth allowance rather than verified individual parts.

Most important distinction

A result can be internally consistent in the 2D point-mass model without being a validated prediction of the physical drone. The report should separate calculated quantities, assumptions, and unvalidated hardware behaviours.

1. Mission-energy consistency

This is the clearest remaining numerical synchronisation task. The final-resolution PDF used an earlier MATLAB run's phase-energy values, while the most recent CSV export contains newer values. The energy ceilings themselves are stated as 207 Wh without towing and 253 Wh with towing; the discrepancy is between the underlying run values and the written complete-mission estimates.

1.1 Values in the latest CSV export

Scenario	Energy in simulated phases	Sizing ceiling	Status in CSV
Attack / no tow	29.85 Wh	207 Wh	Below ceiling; full mission not validated
Defense / no tow	31.74 Wh	207 Wh	Below ceiling; full mission not validated
Tow required	84.73 Wh	253 Wh	Below ceiling; full mission not validated
Abort before launch	0 Wh	Not assessed	Energy compliance not applicable

1.2 Values and method in the resolution PDF

The resolution PDF used 36.84 Wh as the worst-case no-tow simulated-phase value and 95.68 Wh for towing. It added 9.7 Wh for unmodelled no-tow phases and 18.8 Wh for tow phases, producing approximately 46.5 Wh and 114.4 Wh. Those arithmetic results are correct for the older inputs, but they do not match the latest CSV run.

Case	Latest phase-energy value	PDF buffer	Revised additive estimate	Ceiling
Attack no-tow	29.85 Wh	9.7 Wh	39.55 Wh	207 Wh
Worst no-tow (defense)	31.74 Wh	9.7 Wh	41.44 Wh	207 Wh
Tow required	84.73 Wh	18.8 Wh	103.53 Wh	253 Wh

Derived estimates above are simple sums of the latest phase-energy values and the buffers proposed in the resolution PDF. They are not a newly executed continuous mission simulation and must remain labelled as estimates.

1.3 Questions for the guide

- Should the report replace the old 46.5 Wh / 114.4 Wh figures with estimates based on the latest CSV (41.44 Wh worst no-tow / 103.53 Wh tow), or should a new run be made before reporting complete-mission estimates?
- Should MATLAB include launch spin-up, avionics/radio draw, post-release descent/landing, and tow settling as explicit phases, so energy is integrated continuously from launch to landing?
- Please confirm the basis of the 18.8 Wh tow buffer. The stated components are 5.0 Wh launch spin-up, 12.9 Wh loaded descent/settling, and 0.8 Wh avionics; these rounded entries sum to 18.7 Wh. Is the difference due to unrounded underlying values, or should the displayed subtotal be corrected?
- Please confirm the loaded-descent power assumption behind 12.9 Wh. If hover power is scaled linearly from 1.56 kW at 18 kg to 23.5 kg, the estimate is about 2.04 kW; over 20 s that is about 11.3 Wh, not 12.9 Wh. Is 12.9 Wh an intentional conservative allowance based on a different power assumption?
- Is 253 Wh the approved tow sizing ceiling? If yes, please confirm its derivation from the base mission and the 60 s loaded tow allowance, with the 20% demand margin stated explicitly.

Requested outcome

Approve a single energy methodology and a revised pair of numbers: (a) energy actually integrated by the scenario model, and (b) complete-mission estimate including named non-trajectory phases and reserve. Keep "sizing ceiling" distinct from "predicted mission consumption."

2. Closing-speed calculation and guidance audit

The latest CSV includes each target's velocity components and heading. It reports closing speeds of approximately 25.87 m/s (attack), 26.98 m/s (defense), and 28.29 m/s (tow). Those values are not automatically wrong: closing speed is the rate at which range decreases along the line of sight, not necessarily the difference between the two speed magnitudes. However, the latest CSV does not contain enough exported state to reproduce the calculation independently.

2.1 Target inputs currently recorded

Scenario	Target velocity X	Target velocity Y	Target speed	Heading
Attack	14 m/s	5 m/s	14.87 m/s	19.65 deg
Defense	27 m/s	0 m/s	27.00 m/s	0 deg
Tow required	12 m/s	0 m/s	12.00 m/s	0 deg
Abort	-20 m/s	0 m/s	20.00 m/s	180 deg

2.2 Definition to verify

Let r = p_target - p_interceptor, R = norm(r), r_hat = r/R, and v_rel = v_target - v_interceptor. Then the range rate is dR/dt = dot(r_hat, v_rel). Define positive closing speed as V_close = -dR/dt = -dot(r_hat, v_target - v_interceptor). This is the line-of-sight component of relative velocity. It differs from norm(v_target - v_interceptor), which is relative-speed magnitude.

Example of the missing audit trail

To recompute the reported closing speed at the candidate time, the CSV needs the interceptor position and velocity, target position and velocity at the same simulation sample, the line-of-sight unit vector, and the calculated radial range rate. Exporting only the target velocity and the final scalar closing speed is not enough for independent verification.

2.3 Questions for the guide

- Please confirm that the code uses line-of-sight closing speed (negative range rate), not relative-speed magnitude, and that the sign convention is documented.
- Please review whether the scenario headings and target velocity components represent the intended attack, lead-point defense, and tow geometries.
- Can the next CSV export include interceptor Vx/Vy, target Vx/Vy, positions at the capture-candidate sample, LOS unit vector, range rate, and independently recomputed closing speed?
- Are the reported closing speeds (about 25.87 / 26.98 / 28.29 m/s) simply kinematic outputs, or is there a separate acceptable closing-speed envelope for the hypothetical net/capture mechanism? The current point-mass model does not validate capture at these speeds.

2.4 Terminal guidance plots

The resolution document states that near-straight trajectories can be expected when mode selection and lead-point geometry create a near-collision course before proportional navigation dominates inside the blend range. This is plausible, but visual straightness alone does not verify correct PN implementation. A small numerical audit should report line-of-sight rate, commanded lateral acceleration, applied acceleration limit, and miss/range at the candidate event.

Requested outcome

Approve the precise closing-speed definition and the additional CSV fields needed to validate it independently. Confirm whether the current near-straight path is sufficient for the report, or whether a separate deliberately offset test case is needed to demonstrate the PN correction.

3. Forward-flight propulsion margin

The final-resolution PDF performs a sensitivity check using assumed thrust de-rating of 15%, 20%, and 30% relative to the total static thrust. It explicitly says these percentages are bounds for sensitivity analysis, not measurements of the selected G28x9.2 CF propeller. The current simulation should not present them as verified performance data.

Assumed thrust de-rating	Available total thrust	Margin at 18 kg / 25 m/s	Margin at 23.5 kg / 25 m/s
15%	365.2 N	1.93x	1.52x
20%	343.7 N	1.81x	1.43x
30%	300.8 N	1.59x	1.25x

Questions for the guide

- For hackathon-level evidence, is the 15-30% de-rating sensitivity table enough, provided it is clearly labelled as assumed?
- Should the MATLAB model use a conservative 30% thrust de-rating for a single margin check at 25 m/s, or should it include a speed-dependent thrust model? If speed-dependent, what data source or simplifying law should be approved?
- Should the 4.5 m/s^2 cap remain as a conservative commanded-acceleration limit? The note derives larger theoretical static-available acceleration but acknowledges that static thrust does not validate forward-flight dynamics.

Interpretation

The sensitivity table suggests the calculated margin remains above 1.0 under the stated assumptions, but the forward-flight de-rating is unmeasured. The report should say “margin survives the assumed sensitivity range,” not “forward-flight performance has been verified.”

4. Mass ledger and CAD freeze

The revised design baseline uses 18.00 kg nominal mass and 23.50 kg post-capture mass. The listed component subtotal is 14.78 kg, with a 3.22 kg integration/growth allowance to reach the nominal mass. This closes the arithmetic, but not the component-level mass verification.

Item	Current entry	Status / effect
Listed component subtotal	14.78 kg	Ledger subtotal; some component masses still require verification.
Integration/growth allowance	3.22 kg	Named allowance; should not remain an unexplained gap at physical design freeze.
Nominal mass	18.00 kg	Simulation baseline.
Post-capture mass	23.50 kg	Simulation baseline for tow branch.
Battery	12S, 8 Ah, about 1.97 kg	Physical mass, size, mount and usable-energy assumptions must agree with CAD.
Motor / prop / ESC	U8 Lite KV100 / G28x9.2 CF / Alpha 60A 12S	Propeller and ESC masses are identified as not independently verified in the resolution note.

Questions for the guide

- Can the 3.22 kg allowance be accepted as a provisional hackathon budget, or must a component-by-component bill of materials close before CAD freeze?
- Which motor, propeller, ESC and battery dimensions/masses should be treated as authoritative for mounting, arm layout, overall span and centre-of-gravity checks?
- Should the captured-payload/net/parachute mass and envelope be rechecked before freezing the frame and payload bay?

5. Capture/recovery and scenario-boundary assumptions

5.1 Capture and parachute recovery

The resolution note correctly leaves physical capture and recovery unvalidated. The trajectory model can mark a geometric candidate in the assumed 5-15 m range, but it does not model the net opening, mesh interaction with the target, capture success probability, parachute opening shock, canopy dynamics, swinging or tumbling of the combined load.

The sizing document's 4 m net-opening diameter and 2.5 m parachute diameter, with an estimated 4.74 m/s descent rate for a 5.5 kg suspended load, are design-target calculations. They do not establish reliable deployment or a harmless landing.

Questions for the guide

- Does the competition deliverable require only geometric intercept and sizing analysis, or is any model/test of net deployment or parachute behaviour expected?
- What exact language should be used to describe the event: "geometric capture candidate" rather than "successful capture"? The current scope does not support a capture probability.
- Can CAD be frozen while capture/recovery validation remains explicitly outside scope, provided the real hardware mass and envelope are confirmed against the mass ledger?

5.2 Keep-out and safe areas

The 300 m keep-out radius and 120 m safe-area radius remain illustrative scenario boundaries because no actual venue or airport layout has been specified. The resolution PDF refers to DGCA outer airport restrictions as context but explicitly warns that the legal no-fly zone and the tactical simulation keep-out boundary are different concepts.

The MATLAB figures should label the 300 m and 120 m radii as illustrative assumptions, not approved operating limits. Safe-area branch logic can be demonstrated without claiming that these are site-approved dimensions.

Questions for the guide

- Are illustrative 300 m and 120 m boundaries sufficient for the design report, or should the team nominate a hypothetical site and draw a site-specific geometry?
- Can we keep site boundaries as configurable scenario parameters rather than CAD constraints?

6. Decisions requested at the guide meeting

Priority	Decision / support requested	Why it matters
1 - Energy	Approve updated phase-energy values and whether to integrate launch-to-landing energy in one continuous run.	Aligns latest CSV, resolution note and sizing claims.
2 - Closing speed	Approve LOS closing-speed definition; review the added state export and independent calculation.	Makes closing-speed values auditable.
3 - Propulsion	Approve sensitivity-only treatment or specify how to model forward-flight thrust de-rating.	Avoids claiming unmeasured performance.
4 - Mass / CAD	Approve provisional 3.22 kg allowance or require a closed bill of materials before CAD freeze.	MTOW drives propulsion, structure, battery and mounting.
5 - Scope limits	Confirm that capture/recovery and site geometry may remain clearly labelled as unvalidated/illustrative.	Keeps report claims aligned with model scope.

Minimum data to export for independent checks

At candidate sample	Why it is needed
Time, scenario ID, event index	Identify the exact sample used for the closing-speed calculation.
Interceptor position (x, y) and velocity (Vx, Vy)	Required to reproduce relative motion and distinguish range-rate from relative-speed magnitude.
Target position (x, y) and velocity (Vx, Vy)	Confirms the target state matches the scenario inputs and plotted path.
Range, LOS unit vector, range rate, closing speed	Allows the closing-speed scalar to be independently recomputed.
Commanded/applied acceleration and active cap	Shows the guidance command and the value after limiting.
Phase label and electrical power estimate	Allows energy to be integrated consistently over launch, flight, tow, and recovery assumptions.

Source documents reviewed

- SKYFORCE - Final Resolution of Remaining Simulation Issues, 10 October 2026, especially Sections 1-5 (forward-flight sensitivity, energy bounds, capture/recovery limitations, site boundaries, next simulation pass).
- SKYFORCE - Resolution of Simulation Issues (Response to Guide-Review Note), 10 October 2026, especially propulsion baseline, mass ledger, energy ceilings and closing-speed caveat.
- Latest MATLAB export: skyforce_simulation_summary(4).csv. Reported scenario energies: 29.854 Wh attack, 31.742 Wh defense, 84.732 Wh tow; target velocity components and headings included; interceptor velocity components at candidate are not exported in the table.

Expected result from the meeting

A short decision record: (1) approved energy methodology and numbers, (2) closing-speed fields and definition, (3) accepted propulsion de-rating treatment, (4) whether the 3.22 kg allowance is acceptable before CAD freeze, and (5) which validation limitations remain explicitly outside project scope.